Topos Theory Seminar

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if you also want to become an organiser, contact me at

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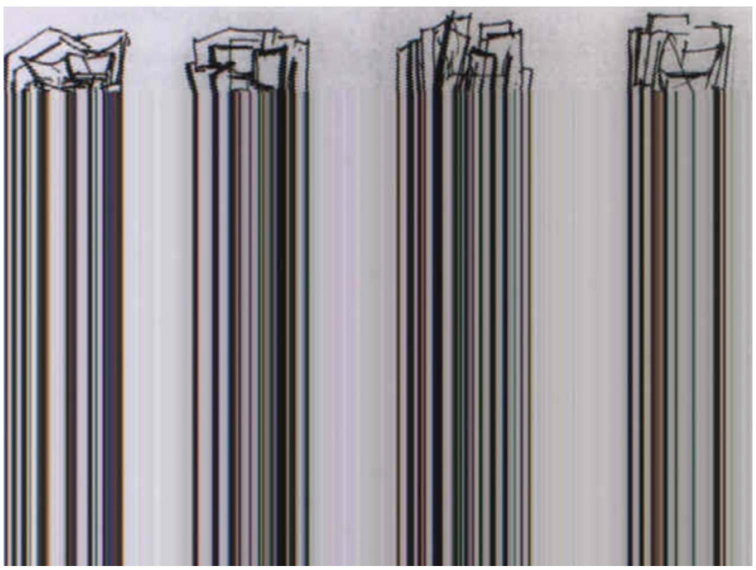

Robert Edward Klippel

Four sculpture motives (or Elephant Is Sleeping, as I would suggest), 1989

Who

Are you a master's or PhD student interested in topology, algebraic geometry, or logic? This seminar is for you. Background in topology and algebraic geometry will be helpful, but not strictly necessary. However, certain familiarity with the basics of category theory will be assumed, so if you are an interested undergraduate student, give it a try!

Why

Some of you attended either the Topology 3 or Algebraic Geometry 1 courses, where we worked with functor categories (simplicial sets) and different types of sheaves, respectively. Topos theory allows us to bring these two contexts under one roof by viewing many explicit constructions from algebraic geometry as special cases of more general constructions in topos theory.

Even if you were not in either of these courses, topos theory is a very interesting subject on its own. Through the seminar, we will discuss various dualities that arise throughout mathematics, how topos theory allows us to understand logic, and why a topos is a *nice place to do math*.

Also, this seminar could be a good accompaniment to the Shape Theory course or the seminar "Cohomology of coherent sheaves" happening this semester.

So, if you want to see the elephant in the room and not just its trunk, this seminar is for you! [1]

What

We will mostly follow the syllabus of the seminar "Sheaves in Geometry and Logic," organised by Stefano Ariotta, Achim Krause, Thomas Nikolaus, and Christoph Schrade some time ago at the University of Münster, which is based on the book "Sheaves in Geometry and Logic" by Mac Lane and Moerdijk.

So, we will use the syllabus of that seminar as a guide. However, there is a lot to cover, and we are not restricted in time, so the main aim of this seminar is understanding rather than coverage. Therefore, some topics in the syllabus may be extended over multiple talks; this will be decided on a case-by-case basis and does not mean that we will go through every single proof.

Once we learn essential basics and techniques, we will be able to discuss more recent developments and applications taking into account the interests of the participants.

List of talks

Talk 1. This talk covers Sections I.1–6. These introduce important categorical concepts, not all of which we have seen before. You should discuss these in relation to (some, not all of) the examples in Section I.1, with focus on the presheaf category $\mathrm{Set}^{\mathbb{C}^{\mathrm{op}}}$. Specifically, introduce subobjects and subobject classifiers as in Sections I.3–4 (including the description in the presheaf category by sieves), discuss the Yoneda Lemma and the perspective on the presheaf category from Section I.5, and explain the notion of exponentials as in Section I.6.

Talk 2. Recall Boolean algebras as in Section I.7. Then define Heyting algebras and discuss their properties following Section I.8, with an emphasis on the categorical perspective taken there. Also cover how to treat quantifiers in this picture (Section I.9).

Talk 3. Section II is rather long, but sheaves are going to be discussed in a more general formal setting in later talks as well. So the purpose of this talk is to familiarize the audience with the notion of sheaves by discussing sheaves on topological spaces, with the goal of establishing the results of II.8. Motivate and introduce the definition of sheaves and discuss Proposition II.1, which characterizes sheaves equivalently through sieves. Feel free to mention

[1] Sketches of an Elephant is a famous book in topos theory. The title refers to the strikingly different aspects of topos theory (such as functorial geometry versus mathematical logic) by alluding to the Indian folklore story of the blind men and the elephant (cf. E. J. Robinson's Tales and Poems of South India), recalled in Johnstone's preface

some of the examples from II.3, but do not go into detail. Explain the perspective of bundles (II.4–6). Section II.8 is the key part of the talk, try to explain these constructions in more detail. End by discussing the functoriality of sheaves as in Section II.9.

Talk 4. Introduce the definition of a Grothendieck topology, and of a basis for it, as in Section III.2, and explain the relation between the two notions. Discuss examples (a)–(f) at the end of the section; in particular, discuss the dense topology and the atomic topology in detail. Define sheaves on a site and prove Proposition III.4.1. Define canonical and subcanonical topologies. Characterize sheaves for the atomic topology. Give the definition of a Grothendieck topos. Give the construction of the associated sheaf functor, and prove Theorem III.5.1.

Talk 5. Prove that every Grothendieck topos is an elementary topos, following Section III.6, to show that it has limits, colimits, and exponentials (leaving out some details if necessary), and Section III.7 to show that it has a subobject classifier. Discuss constant sheaves. Prove Proposition III.8.1, giving an explicit description of the implication operator. Prove Proposition III.8.2, discussing carefully the adjoints of the pullback functor. Conclude saying a few words on the examples at the end of Section III.8.

Talk 6. Give the definition of an (elementary) topos and explain that it is possible to define topoi without any reference to the category of sets whatsoever (as in section IV.1). Explain that the categories of sheaves on a topological space or sheaves on a site give examples of topoi. Discuss some categorical similarities of topoi and the category of sets. Examples of these similarities are given by Proposition IV.1.2 and Theorem V.2.1, which states the existence of exponentials in topoi. Note that in the latter the exponential B^A in a topos is constructed as a certain subobject of the power object of the direct product $A \times B$; as is the case in the category of sets. Finally cover the content of section IV.6. Important results here are Proposition 1 (existence of mono-epi-factorizations in topoi), Proposition 3 ($\mathrm{Sub}(A)$ forms a lattice and pulling back subobjects has a left adjoint) and Proposition 5. In section IV.6 proofs could be left out if there's not enough time.

Talk 7. Explain that the property of being a topos is preserved under taking the slice over an object (this is Theorem IV.7.1). Furthermore discuss Theorem IV.7.2 which states that base-change functors between slice topoi are logical morphisms and admit left and right adjoints. Continue by defining the notions of internal Lattice- and internal Heyting algebra objects in a category (section IV.8). Explain that for an object A in a topos $\mathrm{Sub}(A)$ (resp. $P(A)$) carries the structure of a Heyting algebra (resp. of an internal Heyting algebra object) and that pulling back subobjects defines a homomorphism of Heyting algebras (resp. of internal Heyting algebra objects).

Talk 8. Recall from Proposition IV.6.3 that pulling back subobjects $\mathrm{Sub}(B) \to \mathrm{Sub}(A)$ along a morphism $A \to B$ admits a left adjoint and explain that it also admits a right adjoint (Proposition IV.9.3). Follow the proof of Theorem IV.9.2 to show that this implies the existence of internal left and right adjoints of the pullback morphism $P(B) \to P(A)$ between power objects (also see Proposition IV.9.4). Then explain and prove the Beck-Chevalley conditions for all of these (internal) adjoints (Proposition IV.9.1 and Theorem IV.9.2; also see before Proposition IV.9.4). Proceed by proving that the subobject classifier and power objects in a topos are injective objects (Proposition IV.10.1 and Corollary IV.10.2).

Talk 9. Motivate and introduce the concept of a Lawvere–Tierney topology on a topos as in Section V.1. Especially, discuss carefully the example of presheaves on open sets in a topological space X. Explain the relation to 'closure operators' and how a Grothendieck topology on a category C gives rise to such an operator on presheaves. Introduce the notion of a sheaf for a Lawvere–Tierney topology (as in Section V.2) and spell it out explicitly in the case of a topological space. Follow Section V.2 and explain (roughly) how to see that the category of sheaves for a given Lawvere–Tierney topology is itself a topos.

Talk 10. Give the topos theoretic construction of the 'sheafification' functor as in Section V.3. Try to relate this sheafification functor to the constructions we had earlier in the seminar (using the ++-construction). Also cover the content of Section V.4 proving that for presheaf-categories Lawvere–Tierney topologies are in 1-1 correspondence with Grothendieck topologies and that the notions of sheaves agree. If there is time then discuss several instance of sheaves in this language, for example for the dense and the atomic topology on a presheaf category to illustrate the different viewpoints.

Talk 11. Explain the interplay between internal and external working in a topos as explain in Section V.5. Only briefly state the result of Section V.6 (Theorem 1) and Section V.7 (possibly without giving any details of proof of time does not allow it). Proceed with Section V.9 to explain the 'filter quotient' construction to improve properties of a given topos or more precisely its Heyting Algebra.

When

I suggest meeting this Friday at 10am to assign at least the first couple of talks and to discuss the schedule for the rest of the semester.

Where

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